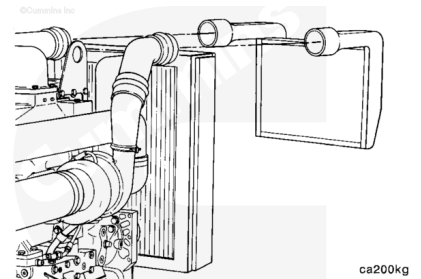


Trainings ▾

Clean

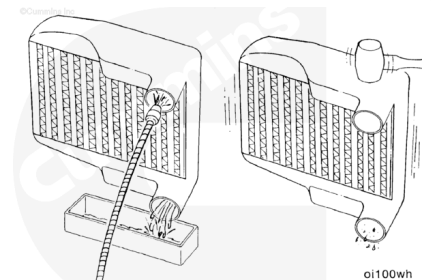
If the engine experiences a turbocharger failure or any other occasion where oil or debris is put into the charge air cooler, the charge air cooler **must** be cleaned.

Remove the charge air cooler piping and charge air cooler from the vehicle. See manufacturer's instructions.



⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

Do not use caustic cleaners to clean the charge air cooler. Damage to the charger air cooler will result.

Flush the charge air cooler internally with solvent in the opposite direction of normal air flow. Shake the charge air cooler and lightly tap on the end tanks with a rubber mallet to dislodge trapped debris. Continue flushing until all debris or oil is removed.

Note : Be sure the tubes are in the vertical direction when flushing.

If the debris can **not** be totally removed from the charge air cooler, the charge air cooler **must** be replaced.

⚠ WARNING ⚠

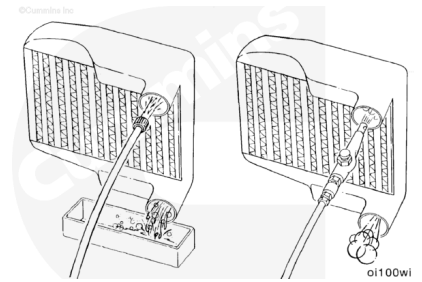
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

The charge air cooler must be rinsed, dried and free of solvent, oil and debris or engine damage will result.

After the charge air cooler has been thoroughly cleaned of all oil and debris with solvent, wash the charge air cooler internally with hot soapy water to remove the remaining solvent. Rinse thoroughly with clean water.

Blow compressed air through the inside of the charge air cooler in the opposite direction of normal air flow until the charge air cooler is dry internally.

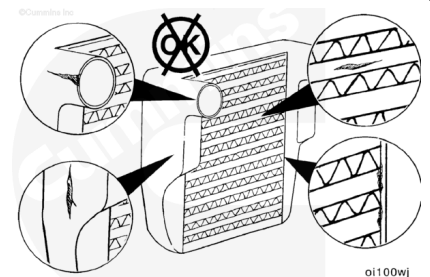


Inspect for Reuse

Inspect the charge air cooler for cracks, holes or damage.

Inspect the tubes, the fins and the welds for tears, breaks or other damage. If any damage causes the charge air cooler to fail the air leak check, the charge air cooler **must** be replaced.

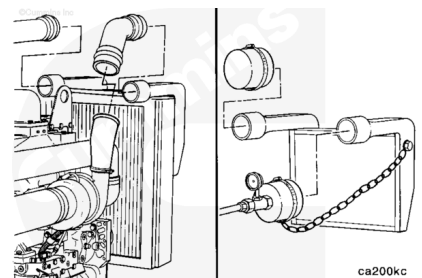
Note : Always clean and inspect the charge air cooler piping and hoses prior to installation.



Leak Test

⚠ WARNING ⚠

To prevent possible personal injury, if either plug blows off during the test, secure safety chains on the test plugs to any convenient capscrew on the radiator assembly. This test must not be performed without securely fastened safety chains.



To check the charge air cooler for cracked tubes or header, remove the inlet and outlet hoses from the cooler. The charge air cooler does **not** have to be removed from the chassis.

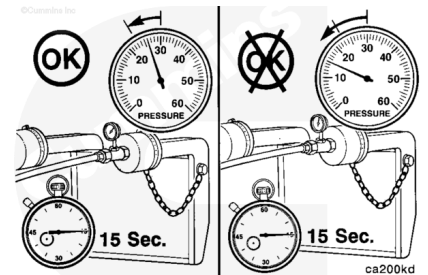
Install a plug or cap over the outlet side of the cooler. Install a pressure gauge and a regulated shop air supply line with a shutoff valve to the inlet side of the cooler.

Apply air pressure to the cooler until the pressure gauge reads a steady 207 kPa [30 psi] of air pressure.

Shut off the air flow to the cooler and start a stopwatch at the same time. Record the leakage at 15 seconds.

If the pressure drop is 34 kPa [5 psi] or less in 15 seconds, the cooler is okay.

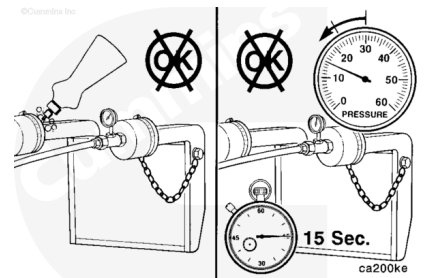
If the pressure drop exceeds 34 kPa [5 psi] in 15 seconds, check all connections again.



Note : If the charge air cooler leaks more than 34 kPa [5 psi] in 15 seconds, it will appear as a major leak in a leak tank.

Determine if the pressure drop is caused by a leak in the charge air cooler or from a leaky connection. Use a spray bottle filled with soapy water applied to all hose connections, and watch for bubbles to appear at the location of the leak.

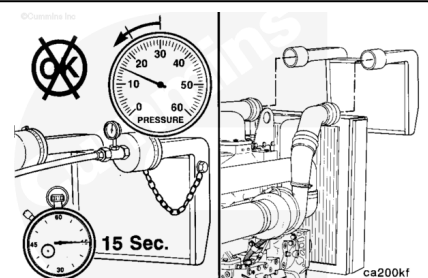
If the pressure drop is caused by a leaky connection, repair the connection and repeat the test. If the leak is within the charger air cooler, repeat the test to verify the accuracy of the pressure drop measurement. Similar pressure drop readings **must** be obtained at least three consecutive tests before the reading can be considered accurate.



Note : Charge air coolers are **not** designed to be 100 percent leak free. If the pressure drop is less than 34 kPa [5 psi] in 15 seconds, then the charge air cooler does **not** need to be replaced

If the pressure drop is greater than 34 kPa [5 psi] in 15 seconds, the charge air cooler **must** be replaced.

See the manufacturer's repair manual for replacement instructions.



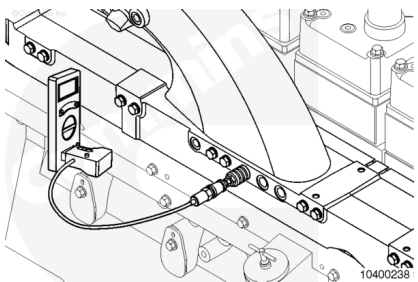
Temperature Differential Test

Inspect the charge air cooler fins for obstructions to airflow. Remove obstructions such as a winterfront or debris. Manually lock shutters in the open position, if equipped.

Install the temperature adapter, Part Number 3164644, along with temperature probe (thermocouple), Part Number 3164498, which are used with digital multimeter, Part Number 3164488, or equivalent components, into the intake manifold at the 1/8-inch NPT near the air horn connection with the intake manifold.

If the engine is equipped with a intake manifold temperature sensor then the INSITE™ electronic service tool can be used to read the temperature instead of a thermocouple. A high performance digital multimeter kit, Part Number 3164489, can also be used.

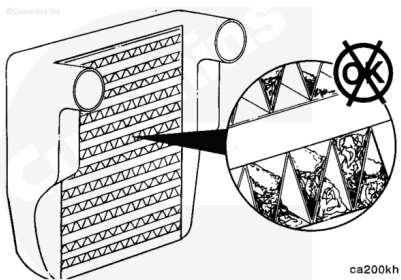
Install another thermocouple at the air cleaner inlet to measure the ambient air temperature.



with Mechanically Actuated Injector

Operate the engine at rated rpm and load. Record the readings on the thermometers.

Record the intake manifold temperature and the ambient air temperature.



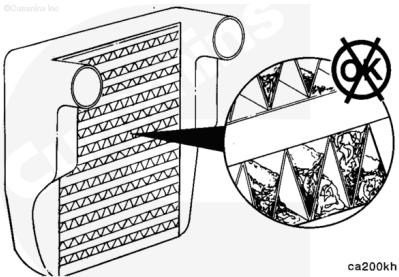
Intake Manifold Temperature	Minus	Ambient Air Temperature	Equals	Differential Temperature
Maximum Differential Temperature 35°C [63°F]				

If the temperature differential is greater than the specifications, check the charge air cooler for dirt and debris on the fins and clean as necessary. If the problem still exists,

check the charger air cooler for debris in the fins or between the charge air cooler and radiator. Confirm full cooling fan engagement.

with Electronically Actuated Injector

Intake Manifold Temperat ure	Minus	Ambient Air Temperat ure	Equals	Differenti al Temperat ure
Maximu m Differenti al Temperat ure 29.9°C [43°F]				



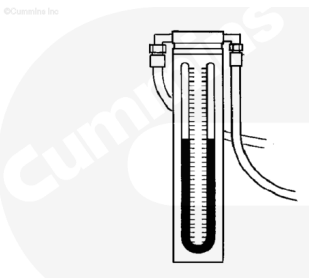
If the temperature differential is greater than the specifications, check the charge air cooler for dirt and debris on the fins and clean as necessary. If the problem still exists, check the charger air cooler for debris in the fins or between the charge air cooler and radiator. Confirm full cooling fan engagement.

Pressure Differential Test

Preferred Method

⚠ WARNING ⚠

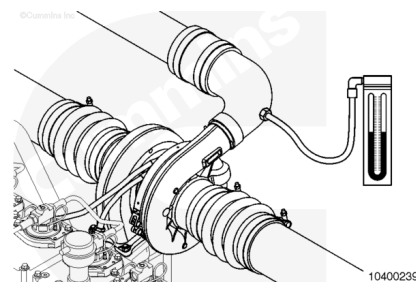
This component contains mercury, a chemical known to some state and federal agencies to cause birth defects or other reproductive harm. Do not dispose. Recycle in accordance with state regulations.



Measure the charge air cooler system pressure drop with a mercury manometer, Part Number ST-1111-3 or digital manometer, Part Number 3164875, or equivalent.

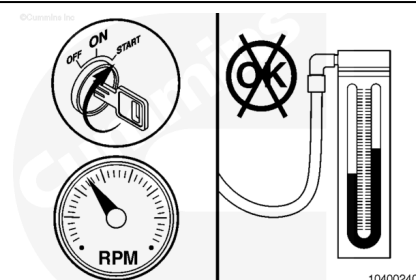
Install one end of a mercury manometer to the fitting in the turbocharger compressor outlet elbow.

Install the other end of the mercury manometer to the intake manifold.



Operate the engine at rated rpm and load. Record the reading on the manometer.

If the differential pressure is greater than 101 mm Hg [4 in Hg], check the charge air cooler and associated piping for plugging. Clean or replace, if necessary.



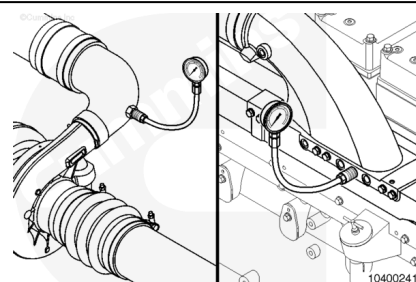
Optional Method

Obtain two pressure gauges, Part Number ST-1273, or equivalent. Check both gauges on the same pressure source at 206 kPa [30 psi] for consistency.

Install pressure gauge, Part Number ST-1273, or equivalent, to the 1/8-inch NPT on the turbocharger compressor outlet elbow.

Install the other pressure gauge in the intake manifold.

If the intake manifold has an intake manifold pressure sensor, then the INSITE™ electronic service tool can be used to measure the pressure.



Operate the engine at rated rpm and load. Record the reading on the manometer.

If the differential pressure is greater than 101 mm Hg [4 in Hg], check the charge air cooler and associated piping for plugging. Clean or replace, if necessary.

